Vehicle Speed Violation Detection using ML Strategies & LPR

Guide : Dr.B.Bhanu Prakash,

M.Tech,PhD,PDF

Professor & Head, Dept. of CSD

~Team SAFE DRIVE CODERS

B.Yasaswini Sai (21JR1A4406)

G.Chaitanya Kumari (4418)

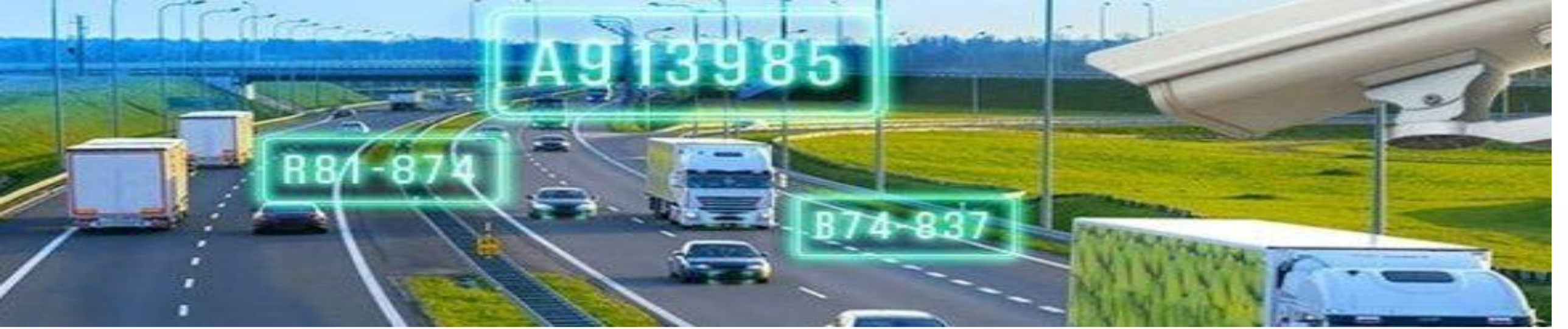
B.Sai Harshitha (4407)

K.Pranavi (4421)

ABSTRACT

Traffic rule violations, such as over speeding and red-light jumping, are primary causes of road accidents, leading to injuries and traffic congestion. This project introduces an innovative Automatic License Plate Recognition (ALPR) system that identifies traffic violations real-time and sends alerts to authorities for immediate enforcement.

The system captures images of vehicles using cameras installed at critical traffic points .The license plate area is extracted from the captured images.The ALPR system then identifies and recognizes the license plate characters and sends alerts to the authorities.



Introduction

Project Overview

This project aims to develop an intelligent system to detect and alert authorities about traffic violations.

Addressing Violations

Traffic violations like speeding and red-light running pose a significant threat to road safety.

ALPR Technology

Automatic License Plate Recognition (ALPR) plays a key role in identifying vehicles and tracking their movements.

Problem Statement

Safety Hazard

Traffic violations are a major contributor to accidents and fatalities, with speeding alone responsible for a significant percentage of these incidents.

Congestion

Rule violations contribute to urban traffic congestion, making commutes longer and more stressful for drivers.

Existing Traffic Violation Monitoring Systems

1

Overview

Current systems largely rely on manual surveillance and static cameras.

2

Technologies Used

Traditional CCTV cameras, manual monitoring, and speed guns.

3

Effectiveness

Typically reactive rather than proactive, with limited real-time enforcement capabilities.

Drawbacks of Existing Systems

Limited Real-Time Monitoring

High latency in detecting and responding to violations.

Human Error

Prone to mistakes in manual operations.

Resource Intensive

Requires substantial human and financial resources for continuous monitoring.

Scalability Issues

Difficult to scale in expanding urban areas.

Project Objectives

Violation Detection

The system uses ALPR to identify speeding and red-light violations.

Real-Time Alerts

Real-time alerts are sent to traffic authorities for immediate enforcement.

Safe Driving

The system promotes safer driving habits and reduces the occurrence of accidents.

Smart City Integration

The system integrates with smart city infrastructure for optimized traffic management.

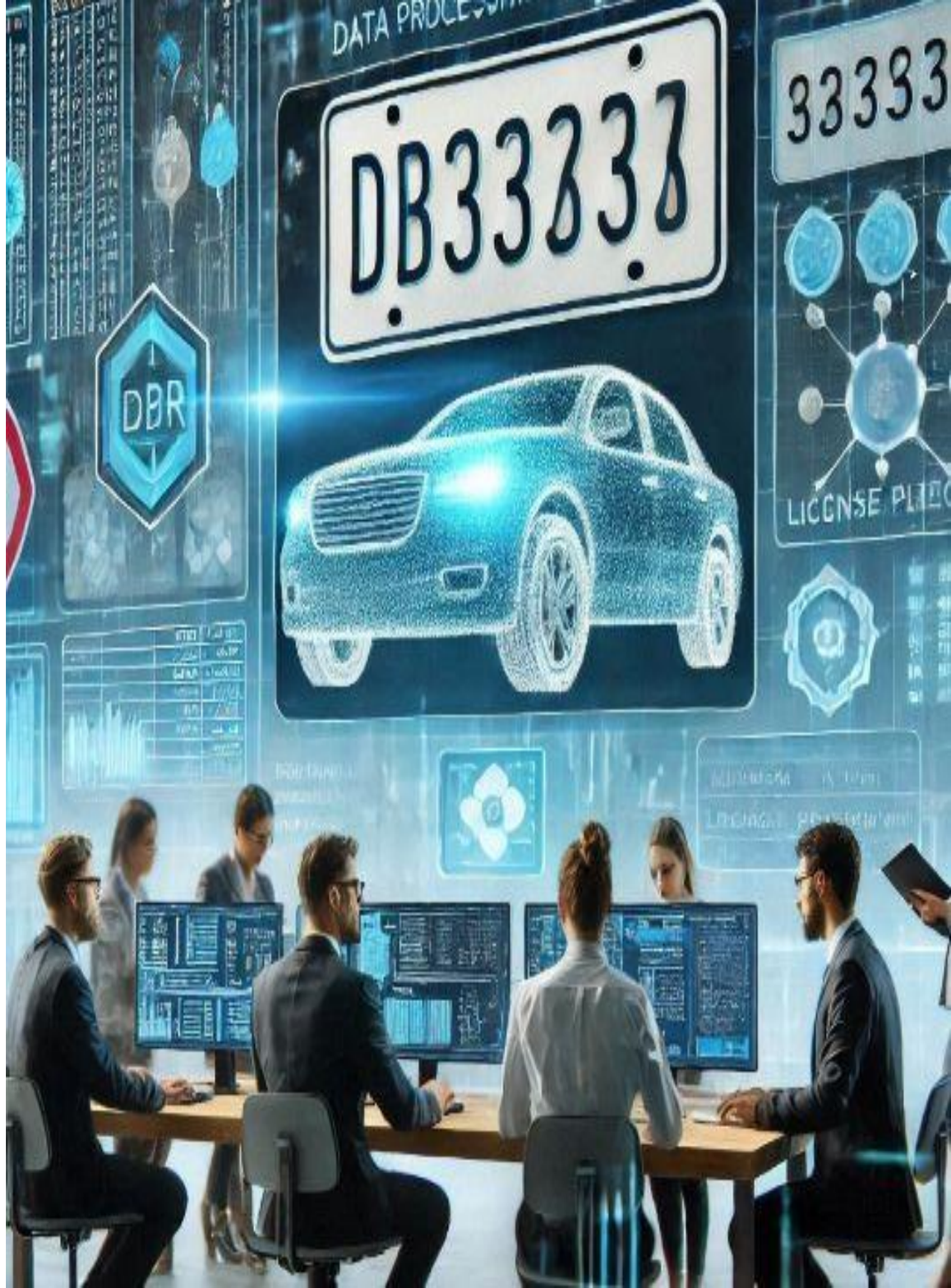
Software Specifications

Recognition Algorithms

The ALPR system utilizes advanced machine learning and computer vision algorithms to accurately detect and recognize license plates under various conditions, including low light and high-speed movement. The software can handle multiple languages and formats of license plates.

Data Handling

Efficient data processing ensures minimal latency. The system captures images, processes them in real-time, and extracts the license plate information. The data is then stored securely in the database for further analysis and reporting.



Hardware Specifications

Cameras

High-definition (HD) cameras with infrared (IR) capabilities for night vision, wide dynamic range (WDR) to handle different lighting conditions, and optical zoom for capturing clear images from a distance.

Placement

Strategically positioned at key traffic points such as intersections, highways, and restricted zones to ensure comprehensive coverage and accuracy.

Durability

Weather-resistant and rugged design to withstand harsh environmental conditions, ensuring reliable performance throughout the year.

VEHICLE SPEED VIOLATION DETECTION

Requirement Analysis

1.Functional Requirements

The system must detect license plates, identify traffic violations, generate real-time alerts, log data, integrate with traffic databases, and support smart city infrastructure for efficient traffic management and enforcement.

2.User Requirements

The system must provide an accurate, reliable, user-friendly interface with remote monitoring, scalability, and automated fine enforcement for efficient traffic violation management.

3.Non Functional Requirements

The system must ensure real-time performance, high accuracy, robust data security, and scalability to handle increased traffic volumes and evolving enforcement needs.

Speed Violation Detection System – Modules Overview

- **Vehicle Detection & Tracking:** YOLO model detects vehicles in real time, while SORT/DeepSORT tracks them across frames.
- **Speed Estimation:** Calculates speed using pixel displacement and real-world calibration, enhanced by CNN-LSTM for accuracy.
- **License Plate Recognition (LPR):** Extracts and recognizes license plates using OCR (Tesseract/LSTM-based models).
- **Violation Detection & Alert System:** Compares speed with legal limits, generates alerts, and stores details like vehicle image and license plate.
- **Data Storage & Reporting:** Logs violations in a database and provides reports for enforcement agencies.

Data Collection and Preprocessing

Traffic footage is collected from CCTV cameras placed at key locations. The collected data includes images and videos of vehicles in motion. Preprocessing techniques enhance image quality by removing noise, adjusting brightness, and focusing on regions of interest. This step ensures that vehicle detection, speed estimation, and license plate recognition work accurately, improving overall system performance.

Feasibility Study

01

1.Technical Feasibility

The system is technically feasible with high-quality cameras, robust computer vision libraries, and seamless integration into existing traffic management infrastructure.

02

2.Operational Feasibility

The system is operationally feasible, offering real-time processing and automation to efficiently manage traffic violations across expanding urban areas.

03

3.Economic Feasibility

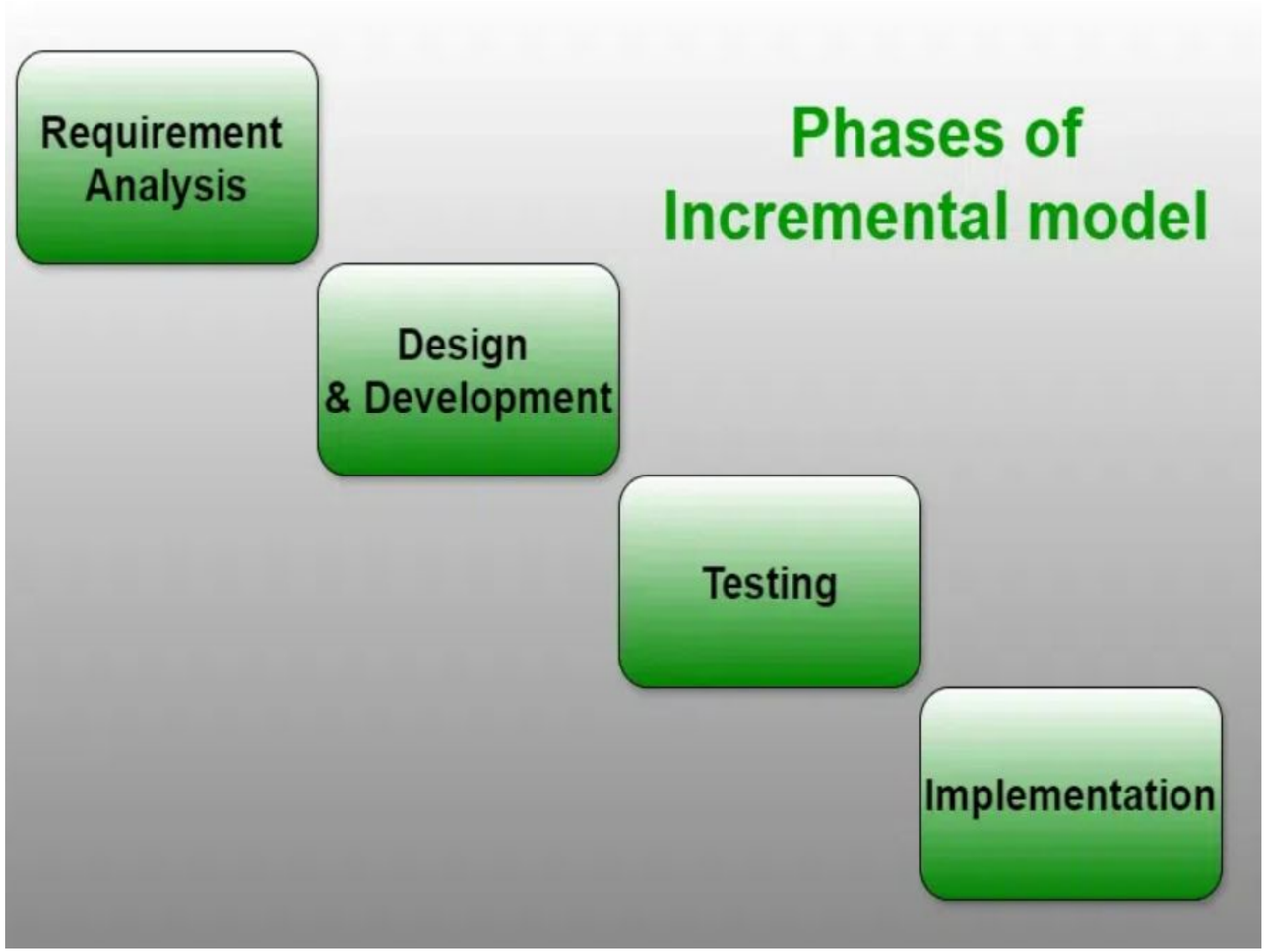
The system is cost-effective, offering long-term savings, revenue generation through automated fines, and sustainable improvements in urban traffic management and safety.

PROCESS MODEL USED

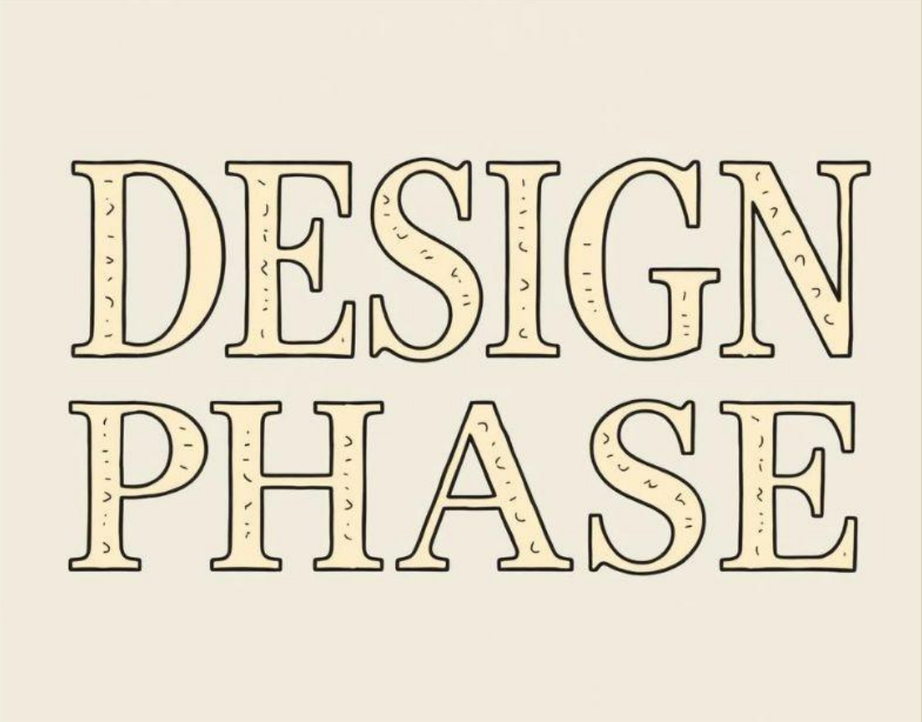
Incremental Model

Ideal for phased development where features are added progressively

1. Initial phase: Basic vehicle detection
2. Next phase: Speed estimation
3. Further enhancement: License plate recognition
4. Final stage: Real-time alert system integration

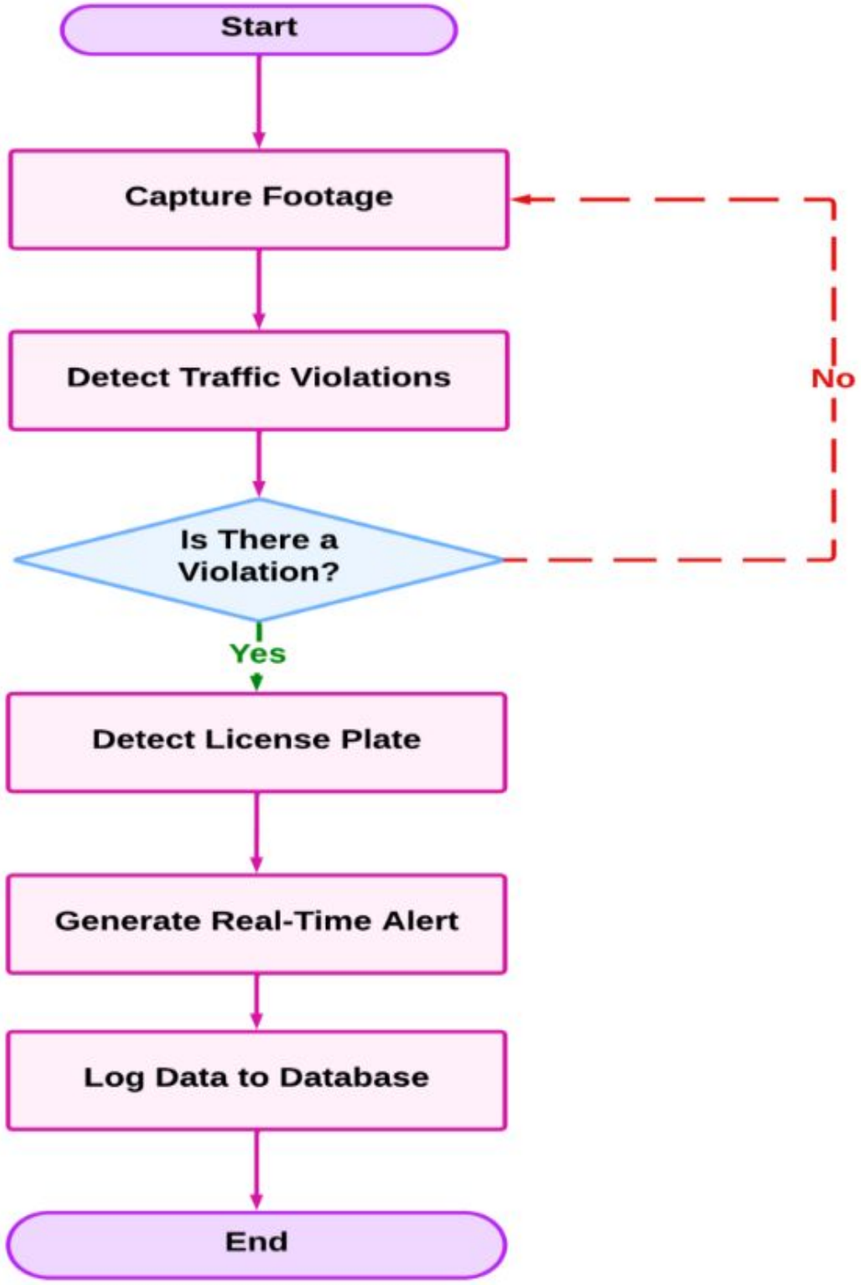


VEHICLE SPEED VIOLATION DETECTION

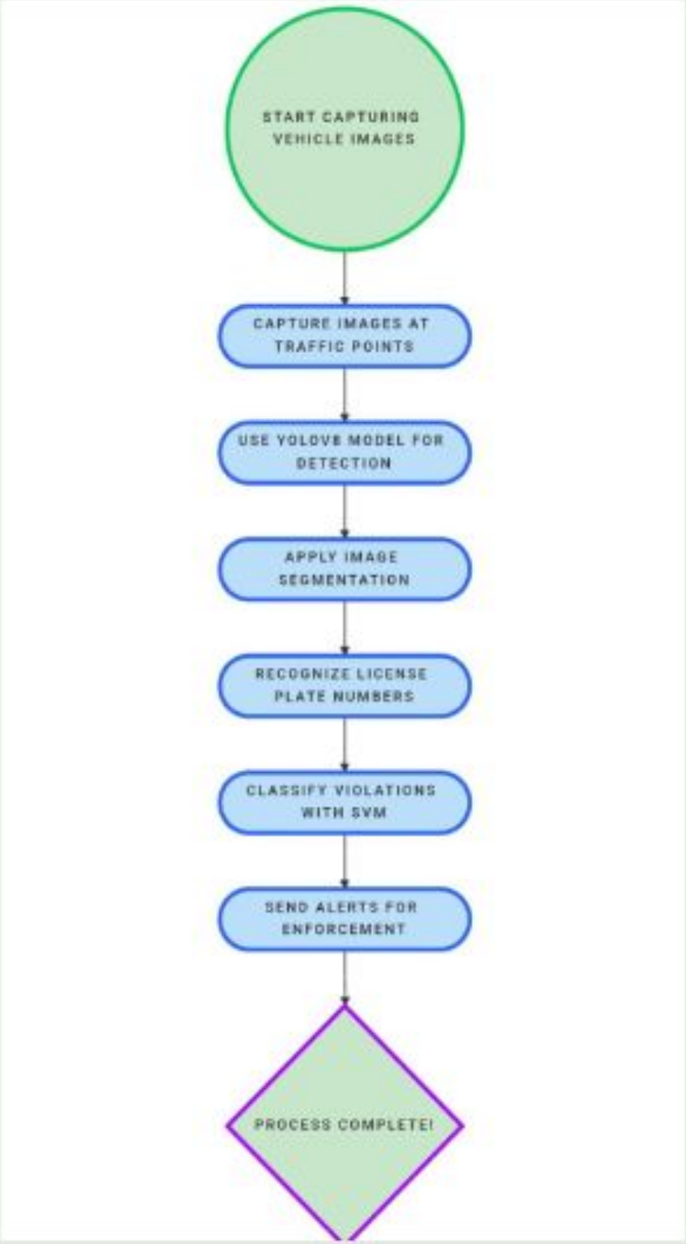


System Architecture

1. Image Capture
2. Image Segmentation
3. Plate Recognition
4. Violation Detection
5. Real-Time Alert



Methodology



- Step 1: Capturing vehicle images at key traffic points.
- Step 2: Using YOLOv8 Model for vehicle detection.
- Step 3: Applying image segmentation to extract license plates.
- Step 4: Recognizing license plate numbers using OCR techniques.
- Step 5: Classifying violations using Support Vector Machine (SVM).
- Step 6: Sending alerts for enforcement.

ACTIVITY DIAGRAM

An activity diagram models the workflow of detecting and managing vehicle violations using footage and plate detection.
Start: The process begins with the system being activated.

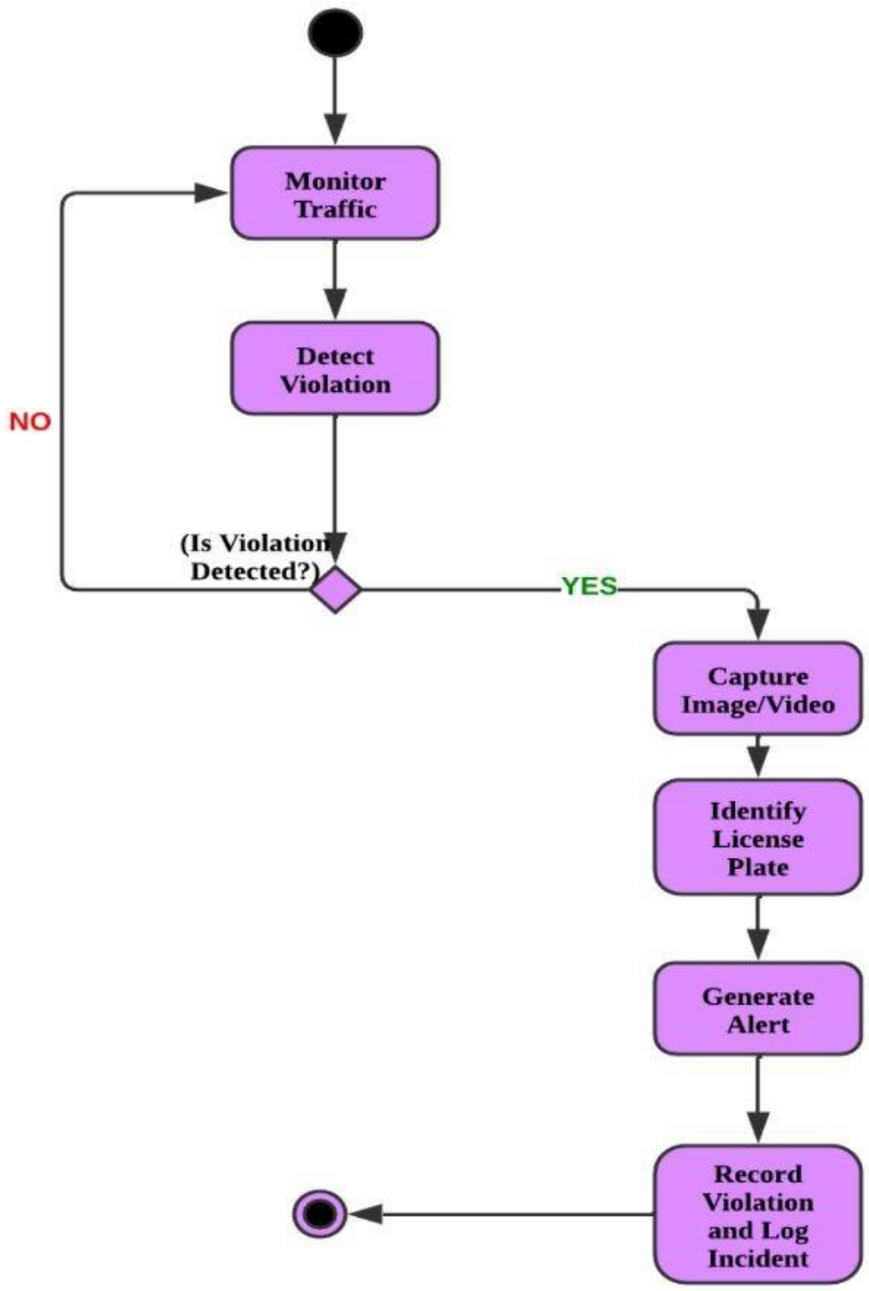
Capture Footage: Cameras capture footage of vehicles in the monitored area.

Detect Plates: The system uses plate detection technology to identify vehicle license plates from the captured footage.

Violation Detected?: A decision point where the system determines if a violation has occurred based on predefined criteria.

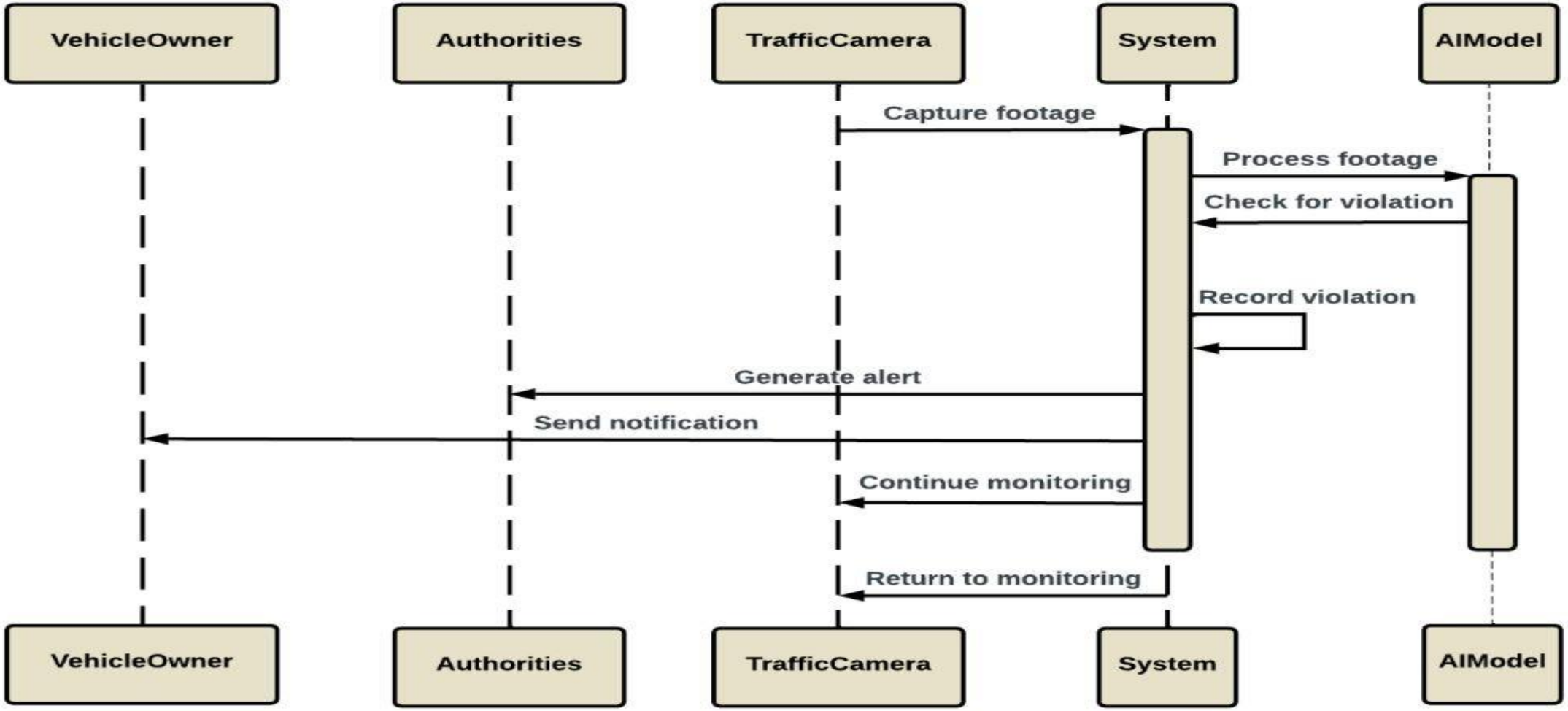
- **If Yes**:**Send Alert**: An alert is sent to the relevant authorities or system administrators.
- **If No**:**End**: The process ends if no violation is detected.

End: The process concludes after either sending an alert or determining no violation occurred.

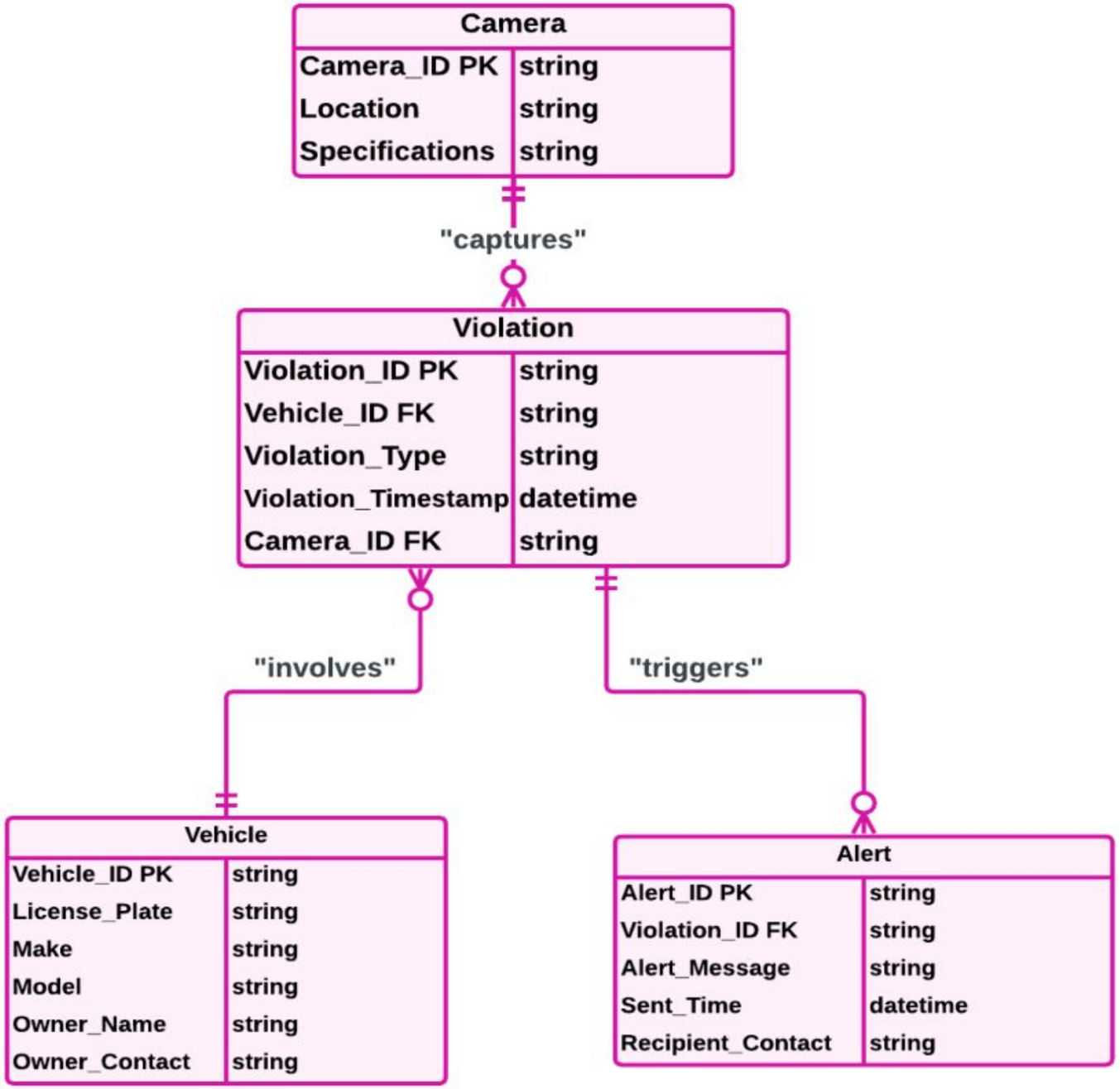


SEQUENCE DIAGRAM

FOR SPEED & VIOLATION ALERT SYSTEM



ER DIAGRAM


Camera
Camera_ID PK string
Location string
Specifications string
"captures"
Violation
Violation_ID PK string
Vehicle_ID FK string
Violation_Type string
Violation_Timestamp datetime
Camera_ID FK string
"involves"
"triggers"
Vehicle
Vehicle_ID PK string
License_Plate string
Make string
Model string
Owner_Name string
Owner_Contact string
Alert
Alert_ID PK string
Violation_ID FK string
Alert_Message string
Sent_Time datetime
Recipient_Contact string

License Plate Detection and Recognition

1 1. Plate Detection

The system employs a deep learning model, such as YOLO, to identify and locate the license plate region within an image or video frame.

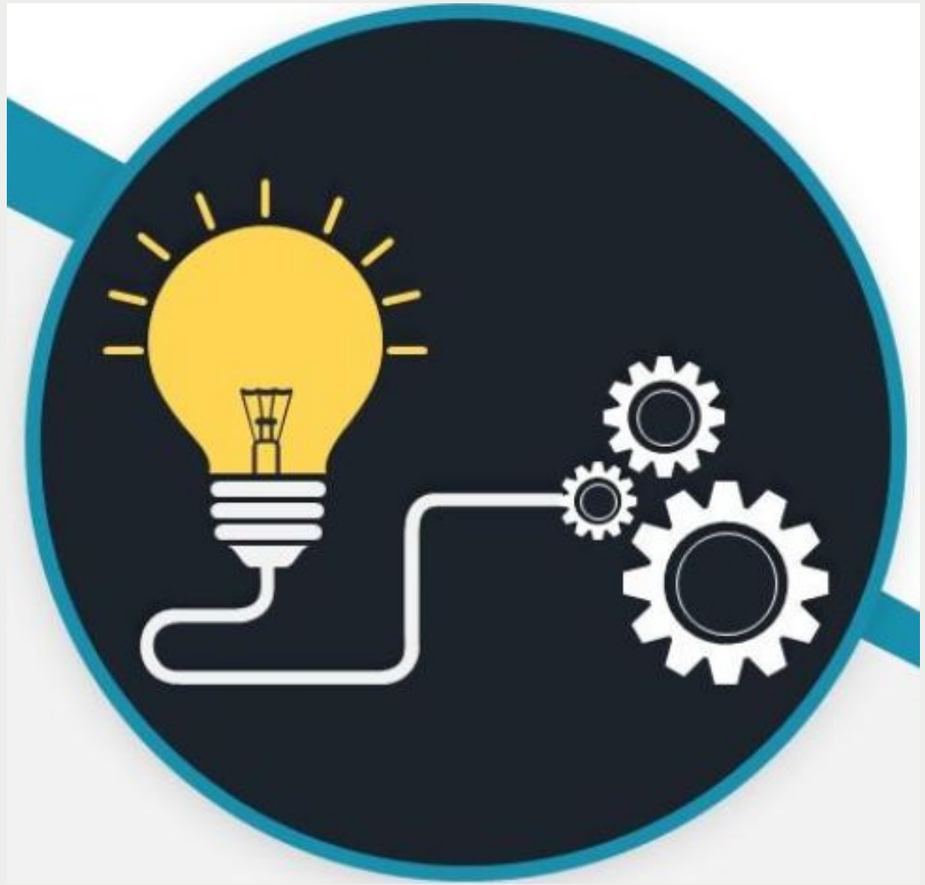
2 2. Image Preprocessing

The detected license plate area is further processed to enhance image quality, remove noise, and prepare it for character recognition.

3 3. Character Recognition

An Optical Character Recognition (OCR) model is applied to recognize the individual characters on the license plate, using algorithms like Tesseract or a specialized deep learning model for character segmentation and classification.

DEVELOPMENT PHASE



Implementation

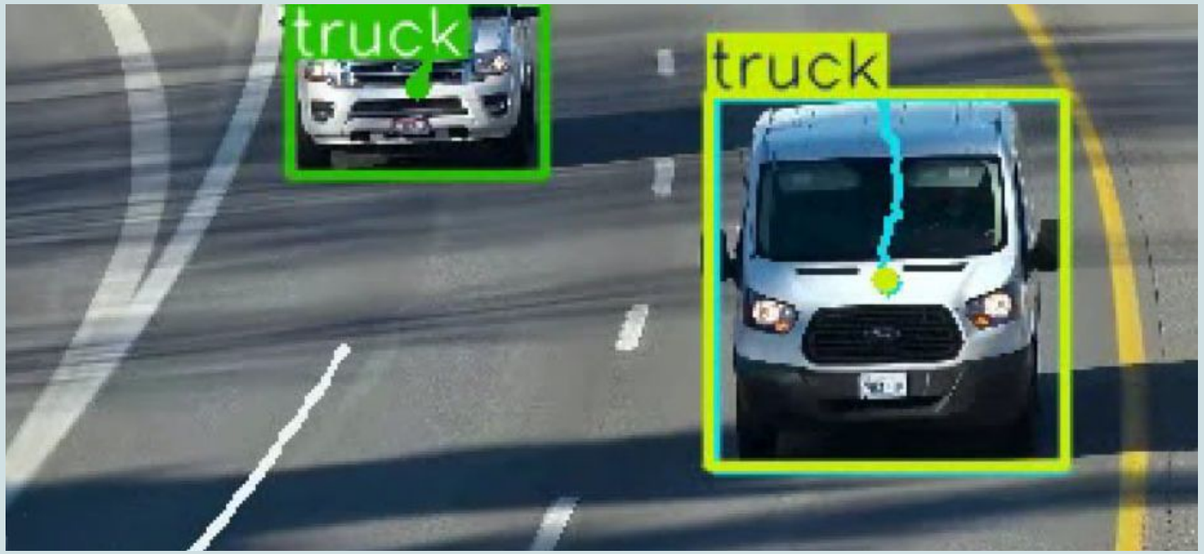
Software Used:

- Programming Language: Python
- Frameworks: OpenCV, YOLO V8
- Libraries: NumPy, Pandas, OpenCV, Matplotlib
- Tools: PyCharm , VS Code

Dataset Used:

- Real-time video footage & pre-labeled vehicle images.

OBJECT DETECTION MODEL



YOLO (You Only Look Once) enables real-time vehicle detection for the speed violation monitoring. It accurately identifies vehicles and provides bounding box coordinates for tracking across frames. Combined with tracking algorithms like SORT, it helps estimate vehicle speed precisely.

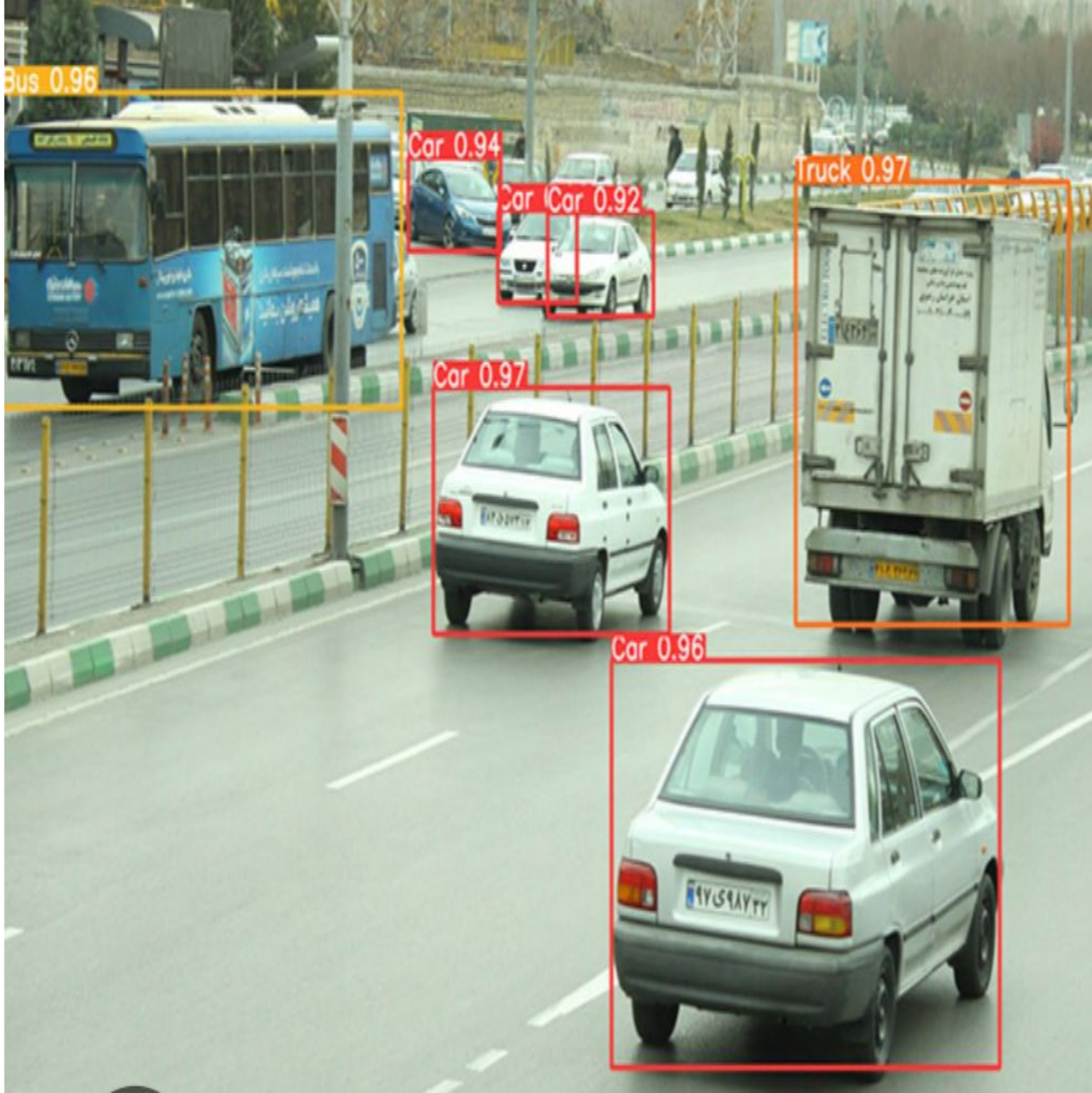
YOLO's efficiency in various conditions makes it ideal for automated traffic surveillance and enforcement.

How YOLO Works?

YOLO divides an image into a grid, where each cell detects objects. It predicts bounding boxes, confidence scores, and class labels. Non-maximum suppression removes duplicate detections, keeping only the most accurate ones. YOLO's deep learning model helps detect vehicles in different lighting and weather conditions. By using pre-trained weights and fine-tuning with custom data, it improves accuracy.

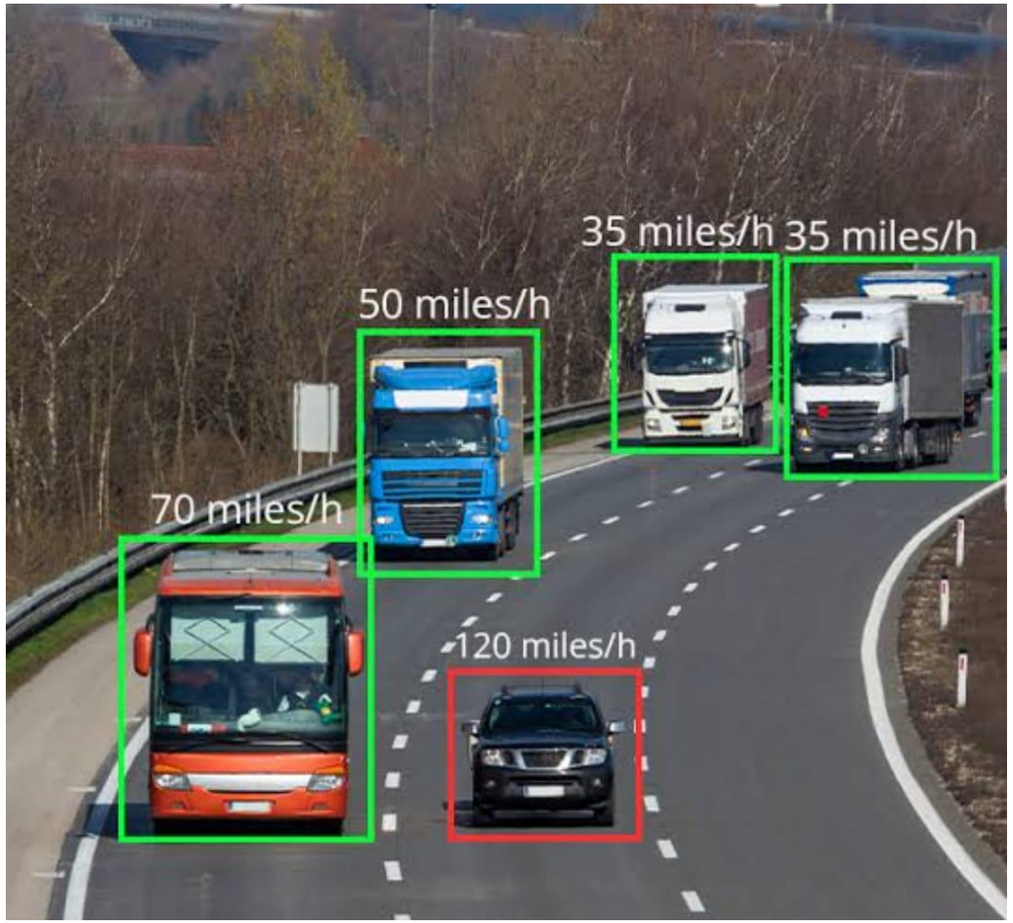
Implementation of YOLO in the Project

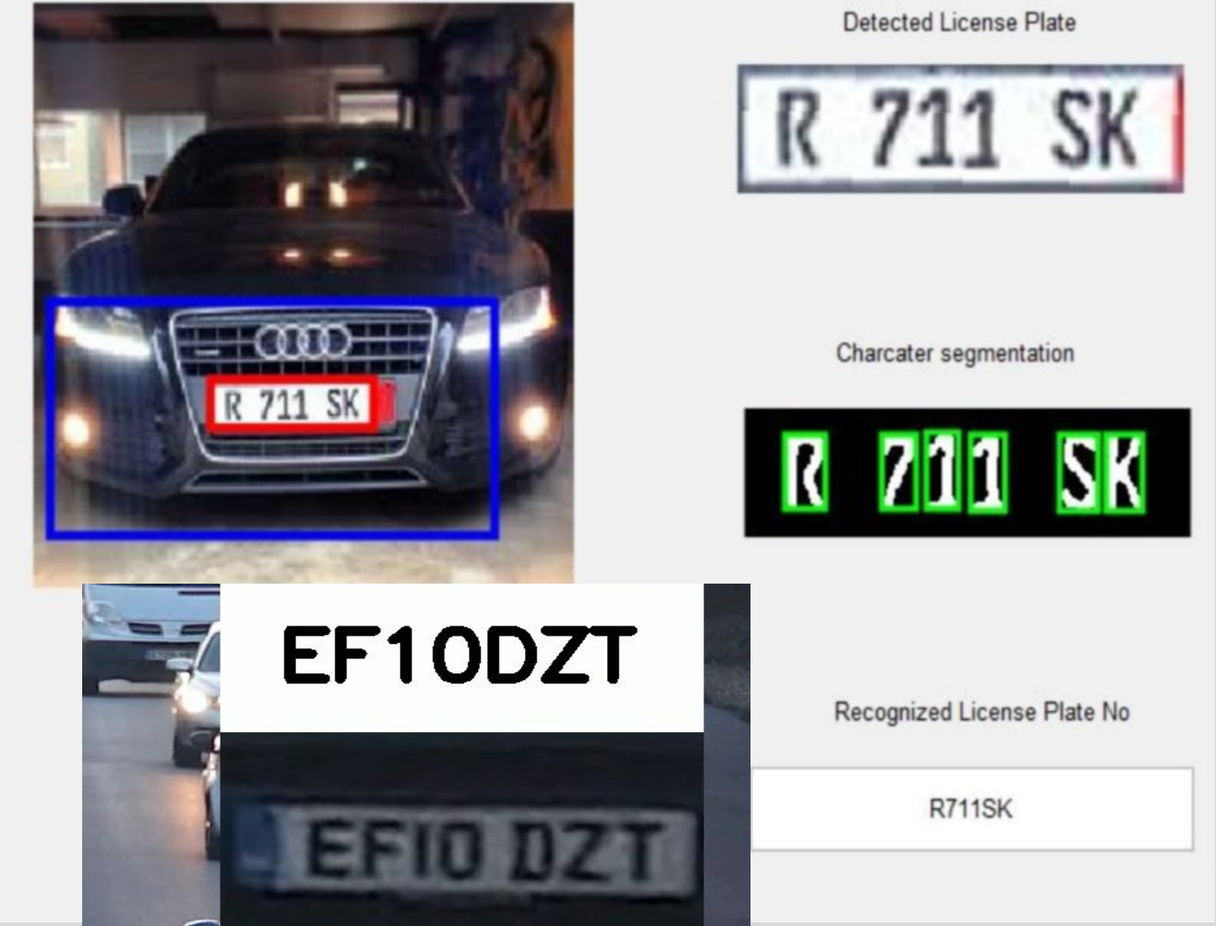
This project uses YOLO for real-time vehicle detection in traffic footage. It identifies cars, trucks, and bikes, then tracks them across frames to estimate speed and detect violations. License plates are also detected and processed using Optical Character Recognition (OCR). YOLO’s fast processing makes it perfect for real-time traffic monitoring



Speed Estimation and Tracking

Speed is calculated by tracking vehicles across multiple frames in a video. Frame-to-frame tracking measures how far a vehicle moves over time to estimate speed. SORT and DeepSORT tracking algorithms help keep track of each vehicle, even if they are briefly blocked or move quickly. These methods improve accuracy and ensure reliable speed monitoring for detecting violations.





License Plate Recognition and OCR

License Plate Recognition (LPR) is a key component of this project, enabling automatic identification of vehicles. Optical Character Recognition (OCR) is used to extract text from detected license plates, converting images into readable characters. However, challenges such as low image quality, motion blur, and varying lighting conditions can affect accuracy. To improve OCR performance, techniques like image preprocessing, noise reduction, and deep learning-based character recognition are applied. By refining detection and enhancing OCR accuracy, the system ensures reliable identification of vehicles for effective traffic monitoring.

CODE IMPLEMENTATION

```python
from ultralytics import YOLO
import cv2

import util
from sort.sort import *
from util import get_car, read_license_plate, write_csv


results = {}

mot_tracker = Sort()

# load models
coco_model = YOLO('yolov8n.pt')
license_plate_detector = YOLO('license_plate_detector.pt')

# load video
cap = cv2.VideoCapture("Traffic Control CCTV.mp4")

vehicles = [2, 3, 5, 7]

# read frames
frame_nmr = -1
ret = True
while ret:
    frame_nmr += 1
    ret, frame = cap.read()
    if ret:
        results[frame_nmr] = {}
        # detect vehicles
        detections = coco_model(frame)[0]
```

1.Main.py

- Vehicle Detection-YOLO V8 model detects vehicles from video frames.
- Support Vector Machine (SVM) used for classification.
- Speed estimation using video processing.
- Violation Detection
- License plate extracted from detected vehicles.
- OCR applied for text recognition.
- Getting the characters of License plate output.

main.py | speed_estimation_1.py | add_missing_data.py | visualize.py

```python
import cv2
from ultralytics import solutions

# Video input and output settings
input_video_path = "car_video_1.mp4"
output_video_path = "speed_estimation.avi"
resize_width = 640  # Desired width of the resized video
resize_height = 360  # Desired height of the resized video

cap = cv2.VideoCapture(input_video_path)
original_fps = int(cap.get(cv2.CAP_PROP_FPS))
video_writer = cv2.VideoWriter(
    output_video_path, cv2.VideoWriter_fourcc(*"mp4v"), original_fps, (resize_width, resize_height)
)

# Initialize SpeedEstimator with vehicle class filter
speed_obj = solutions.SpeedEstimator(
    region=[(0, resize_height // 2), (resize_width, resize_height // 2)],
    model="yolo11n.pt",
    classes=[2, 3, 5, 7],  # COCO classes for vehicles: car, motorcycle, bus, truck
    show=True,
)

while cap.isOpened():
    success, frame = cap.read()
    if not success:
        break

    # Resize the frame
    resized_frame = cv2.resize(frame, dsize=(resize_width, resize_height))
```

2.Speed estimation.py

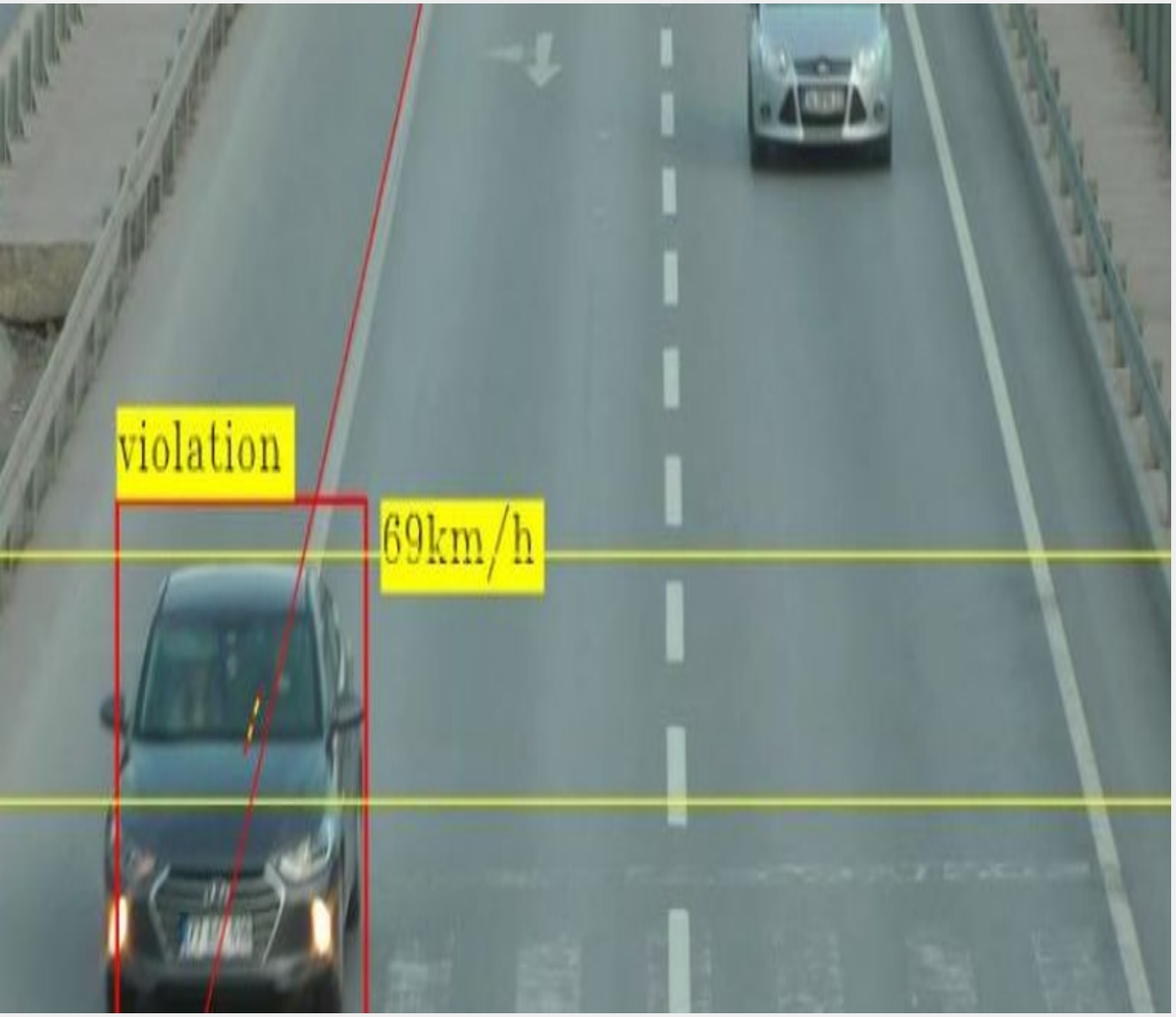
3.Visualize.py

main.py | speed_estimation_1.py | add_missing_data.py | visualize.py

```python
# load video
video_path = 'Traffic Control CCTV.mp4'
cap = cv2.VideoCapture(video_path)

fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # Specify the codec
fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
out = cv2.VideoWriter('./out.mp4', fourcc, fps, (width, height))

license_plate = {}
for car_id in np.unique(results['car_id']):
    max_ = np.amax(results[results['car_id'] == car_id]['license_number_score'])
    license_plate[car_id] = {'license_crop': None,
                             'license_plate_number': results[(results['car_id'] == car_id) &
                                                             (results['license_number_score'] == max_)]['license_number'].iloc[0]}
    cap.set(cv2.CAP_PROP_POS_FRAMES, results[(results['car_id'] == car_id) &
                                             (results['license_number_score'] == max_)]['frame_nmr'].iloc[0])
    ret, frame = cap.read()

    x1, y1, x2, y2 = ast.literal_eval(results[(results['car_id'] == car_id) &
                                              (results['license_number_score'] == max_)]['license_plate_bbox'].iloc[0].replace('[ ',

    license_crop = frame[int(y1):int(y2), int(x1):int(x2), :]
    license_crop = cv2.resize(license_crop, dsize=(int((x2 - x1) * 400 / (y2 - y1)), 400))

    license_plate[car_id]['license_crop'] = license_crop
```

Testing

- Conducted testing on real-world traffic footage.
- Evaluated accuracy, speed estimation errors, and violation detection.
- Performance metrics such as Precision, Recall, and F1-score.



Performance Metrics in Speed Violation Detection

Accuracy – Measures overall correctness in detecting vehicles, estimating speed, and recognizing license plates.

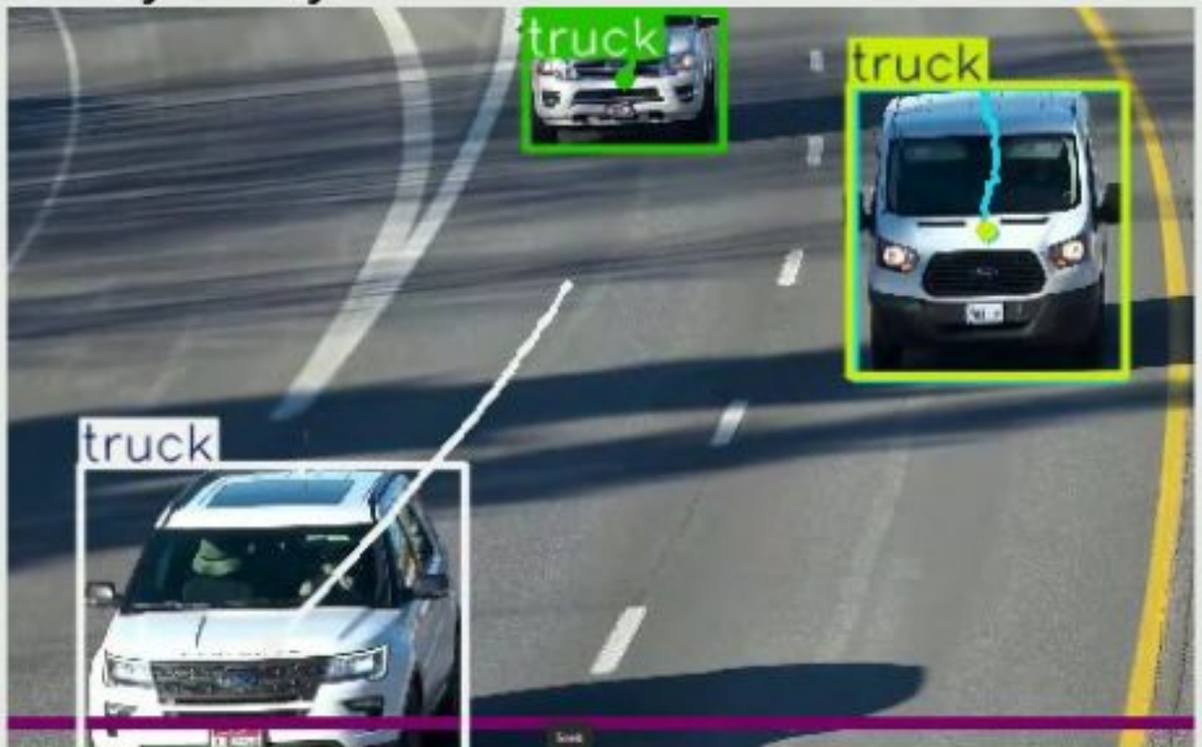
Precision – Ensures detected vehicles and plates are correct, reducing false alarms.

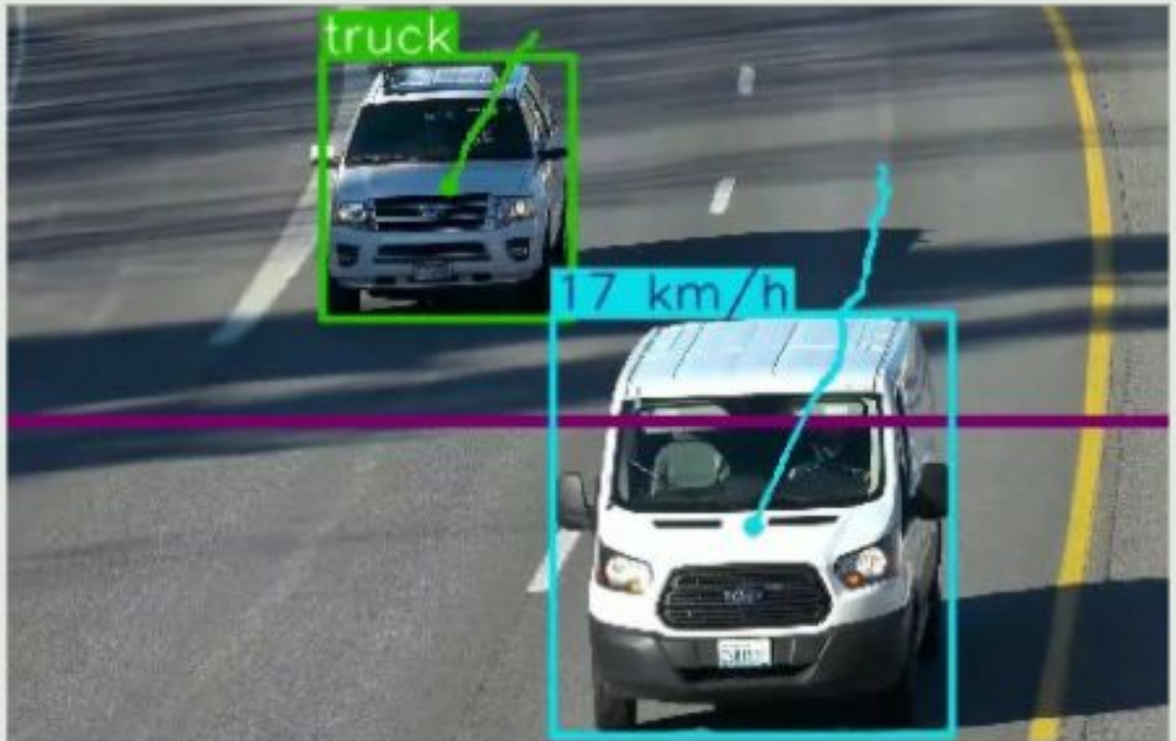
Recall – Checks how well the system finds all vehicles and plates, avoiding missed detections.

F1-Score – Balances precision and recall for reliable and accurate monitoring.

Output Screens (Speed & Vehicle Detection)

The YOLO-based vehicle detection system processes traffic video and identifies vehicles in real-time. Output screens display images with annotated bounding boxes around detected vehicles. The system efficiently tracks multiple vehicles, ensuring accurate localization and monitoring. High-speed detection enhances traffic management and road safety analysis.





Output Screens (License Plate Recognition)

The system extracts license plates from detected vehicles and displays them on output screens. OCR algorithms recognize and convert plate numbers into machine-readable text. It ensures accuracy under varying lighting and plate orientations. This feature supports law enforcement and automated toll systems for efficient vehicle identification.

Output Screens (Violation Detection & Alert System)

- The system detects traffic violations such as overspeeding and red-light breaches in real-time. Output screens display image frames highlighting these violations for accurate enforcement.

- An automated alert system generates reports and logs violation details, including license plate numbers. A CSV file is created to store and manage violation records for further analysis and action.

Advantages of the System

Real-Time Detection

The system provides immediate identification of violations.

Swift Enforcement

Alerts enable prompt enforcement of traffic rules.

Congestion Management

Enhanced traffic flow improves overall congestion management.

Accident Reduction

The system contributes to a significant reduction in road accidents.

Responsible Driving

The system encourages drivers to adopt responsible driving practices.

Applications of the System

1. **Smart Cities**
   Integrated traffic management systems for urban environments.

2. **Highway Management**
   Real-time monitoring and enforcement on highways.

3. **Red-Light Systems**
   Dedicated red-light violation detection at intersections.

4. **CCTV Integration**
   Integration with existing CCTV surveillance infrastructure.

5. **Parking/Restricted Zones**
   Monitoring parking violations and restricted zones.

Expected Results

30-40%

Speeding Reduction

A significant reduction in speeding incidents.

Enhanced

Enforcement Response

Improved response times for traffic enforcement.

Improved

Safety Statistics

Positive impact on road safety statistics.

Conclusion

This project aims to develop a sophisticated Speed and Violation Alert System, harnessing the power of ALPR technology to enhance road safety, promote responsible driving habits, and contribute to the development of smarter and safer transportation infrastructure for the future. The system is poised to play a vital role in reducing traffic violations, accidents, and congestion, making our roads safer and our cities more livable.

Thank you